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Unit 1 - Exploring the Core

PDF Documents

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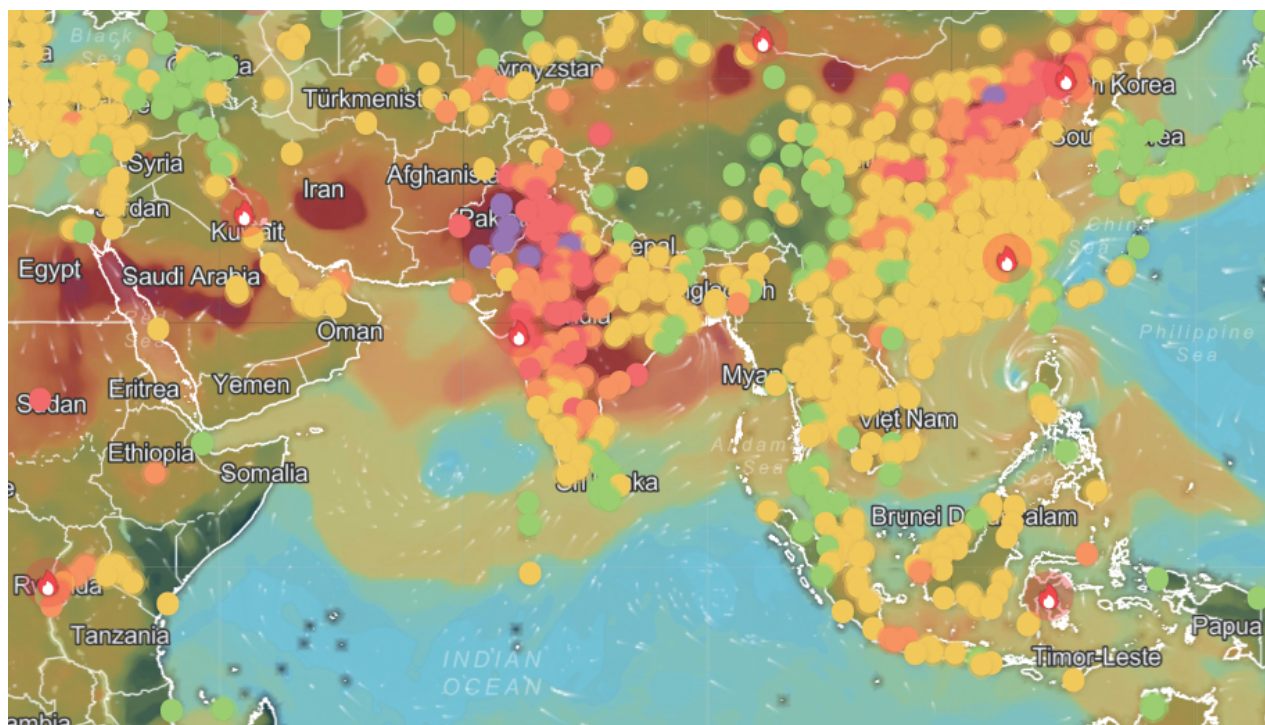


UNIT 1 EXPLORING THE CORE

Topic Investigation

• Global air quality

Global Air Quality Index (AQI⁺) and PM2.5 Pollution



Air quality rankings of major cities around the world

#	CITY	UNITED STATES AQI ⁺
1	 Delhi , India	220
2	 Beijing , China	216
3	 Lahore , Pakistan	194
4	 Kampala , Uganda	164
5	 Mumbai , India	158
6	 Tashkent , Uzbekistan	149
7	 Hanoi , Vietnam	128
8	 Algiers , Algeria	117
9	 Karachi , Pakistan	113
10	 Sarajevo , Bosnia and He...	111

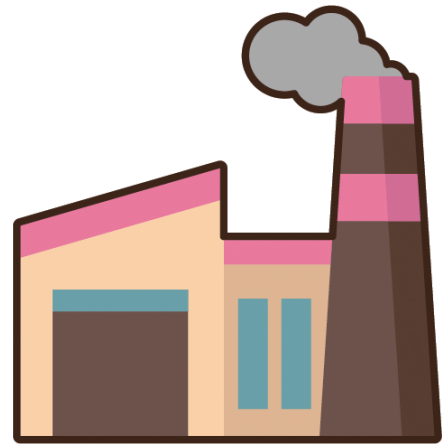
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It is well-known that air pollution, such as fine dust, is severe worldwide. Environmental issues profoundly impact our health and lives, with children and older people receiving warnings to refrain from going out depending on the air quality. Let's find out the causes of environmental pollution, investigate its impact on our health, and find out how to overcome it through technological efforts.

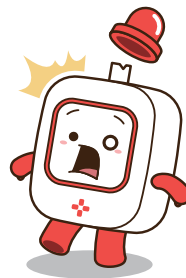
CAUSES OF AIR POLLUTION



Exhaust gas from automobiles and factories: When automobiles are driven, and factories and power plants attempt to produce electricity and products, exhaust gases are produced through fuel combustion. Harmful gases in this exhaust mix with the air and cause pollution. Air pollution is so severe in cities because of heavy traffic and congestion.



Waste and trash: When trash is incinerated, harmful chemicals are released into the air, and certain wastes can decompose, releasing gases into the air and worsening air quality.



Fires and natural disasters: Forest fires, dust storms, and volcanic eruptions release smoke and dust particles.



- **The impact of air pollution on health**

These causes add small particles and chemicals into the air, making breathing difficult and even causing health problems. People with asthma or allergies, children, and the elderly are especially affected. These problems affect not only humans but all animals and plants living on Earth. Acid rain wilts plants, pollutes our water, and even damages growth.



- **Technical efforts to combat air pollution and climate change**

The technologies outlined below address the immediate effects of air pollution and slow the effects of climate change, helping to create a sustainable and healthy planet.



■ Renewable energy

Reduce dependence on fossil fuels, a significant source of pollution, through solar, wind, and hydroelectric power generation.

■ Smart agriculture

IoT sensors monitor soil, water, and crop conditions to reduce waste and use fewer resources through efficient agricultural activities.

■ Climate monitoring with AI and data analytics

AI and IoT devices monitor air quality to minimize damage. They allow cities to issue warnings and take action when pollution spikes.

■ Energy-efficient home appliances

It creates clean air by reducing energy consumption and exhaust gas emissions.





Inquiry & Reflection

Discussion Question 1

What are some things we can do at home or school to help keep the air clean?

Discussion Question 2

How do you think technology, like robots or sensors, can help us make the air cleaner and safer?